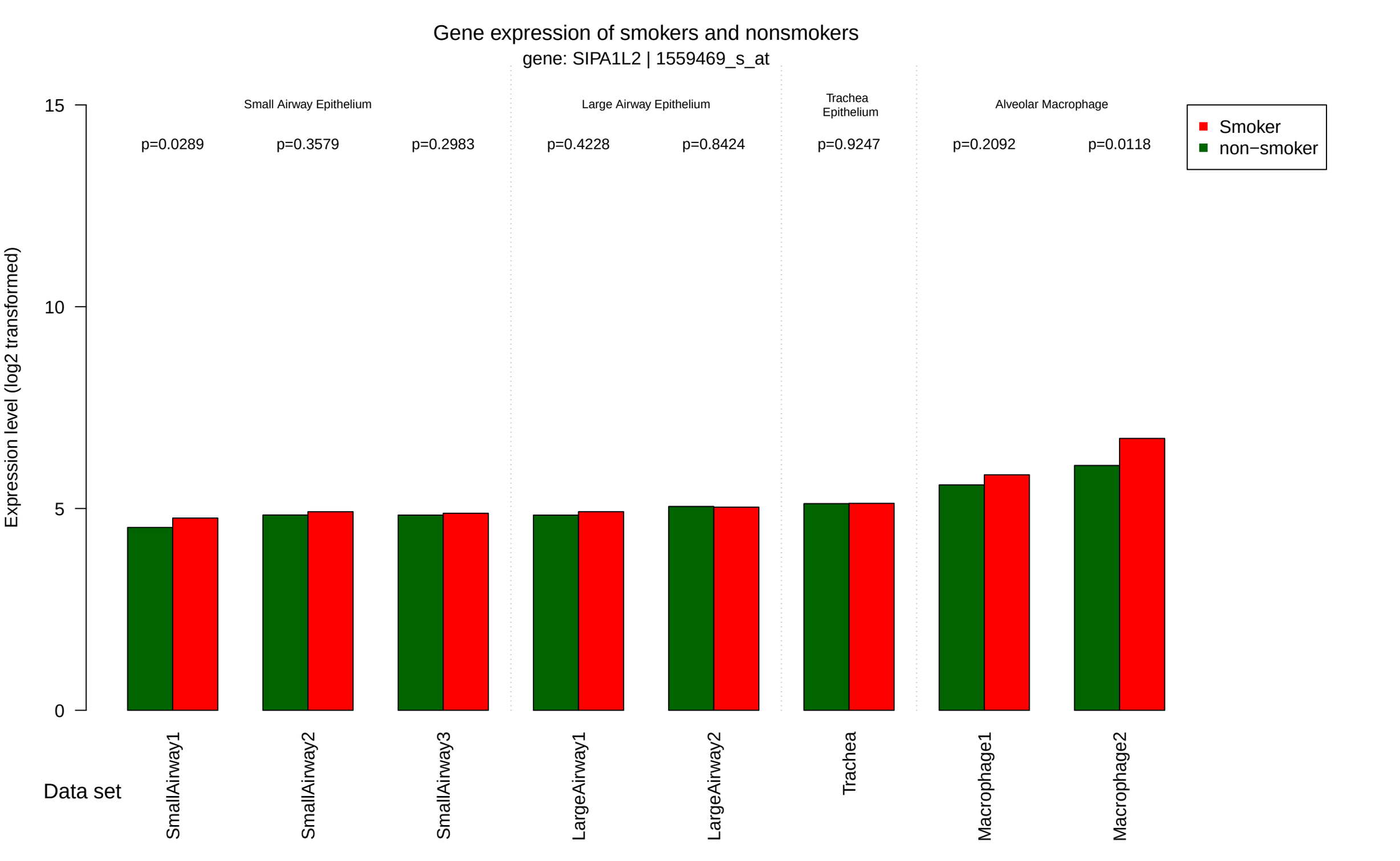
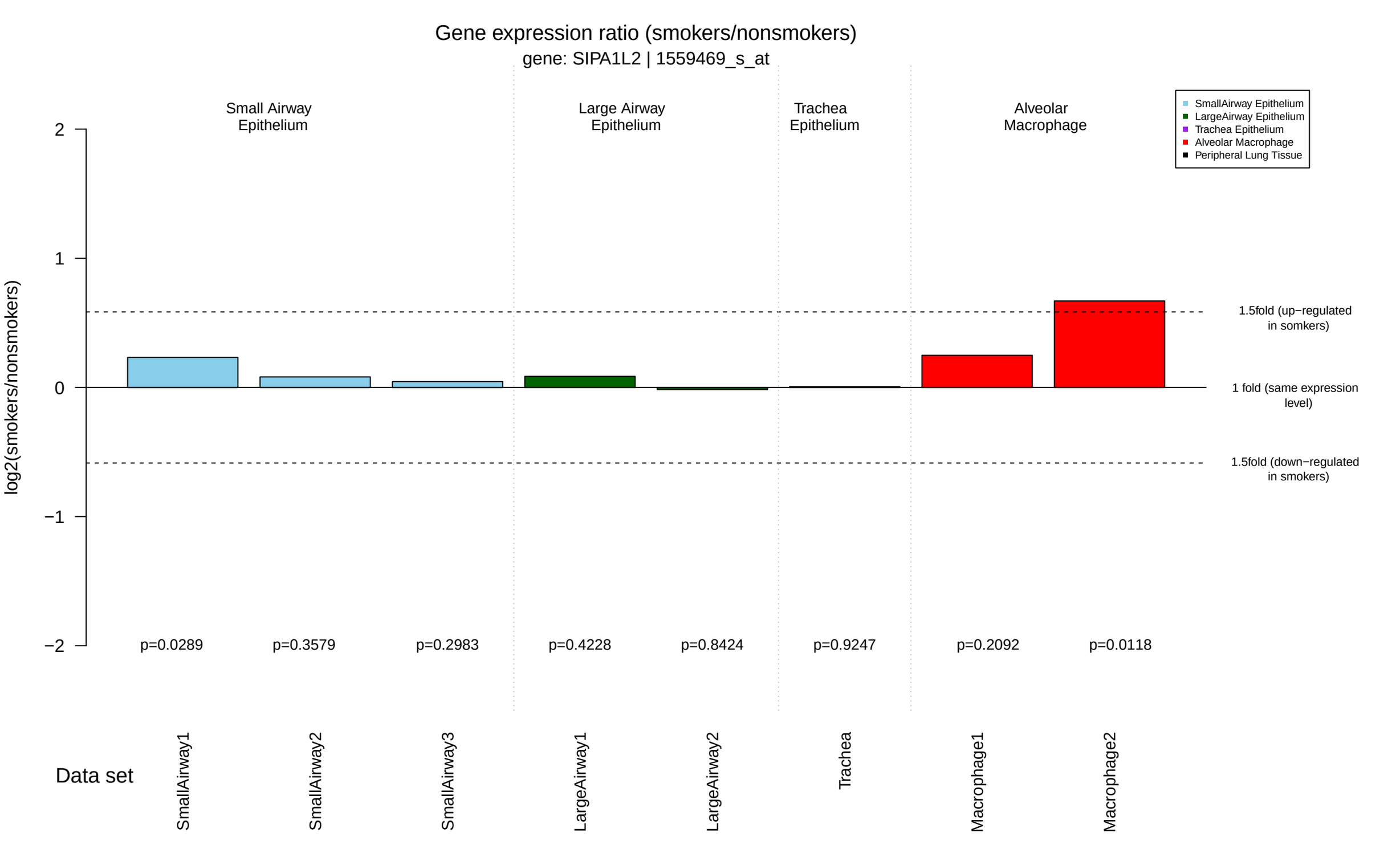
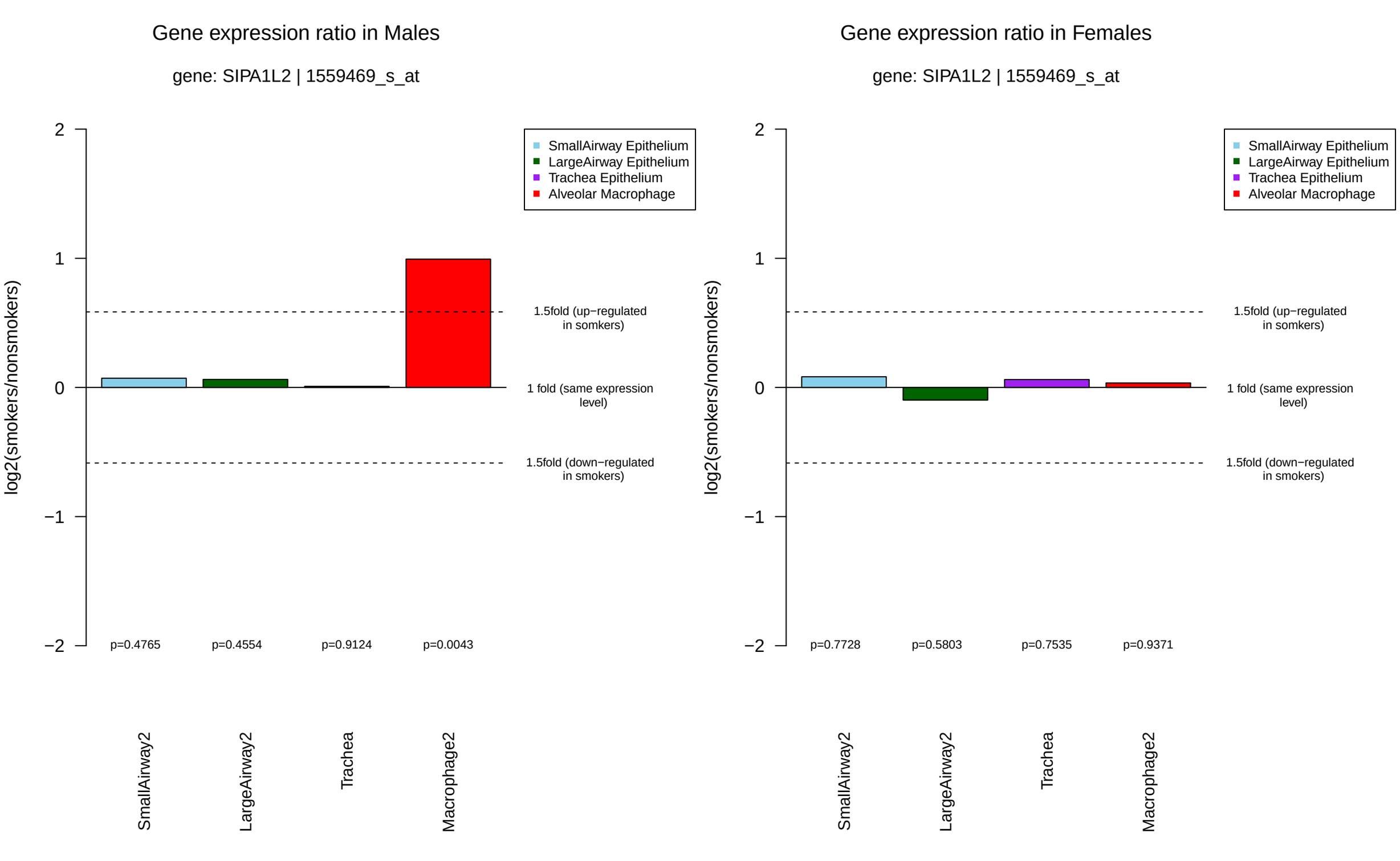
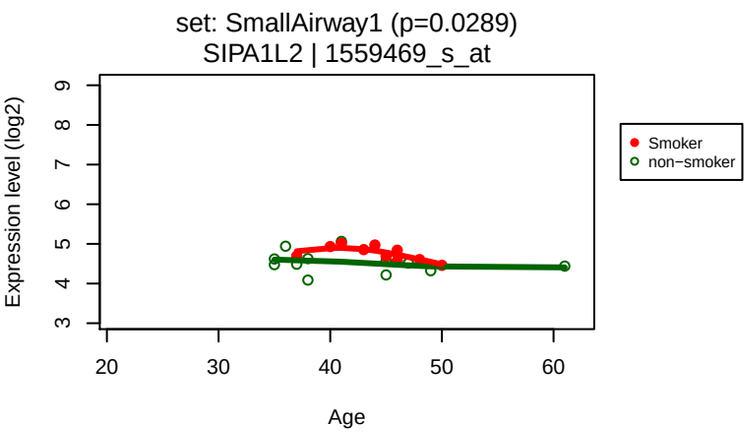
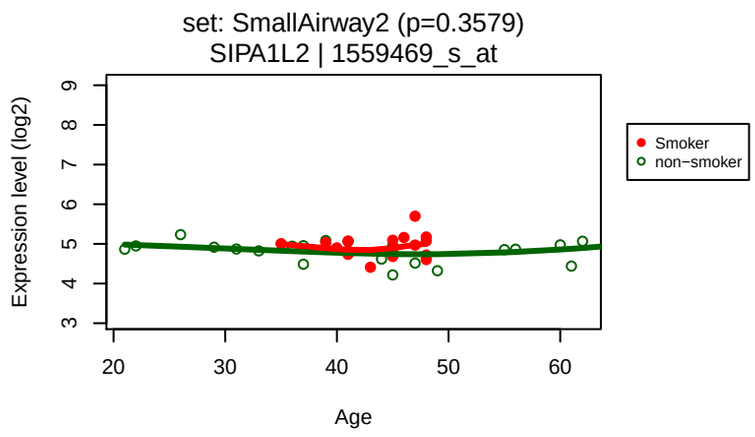
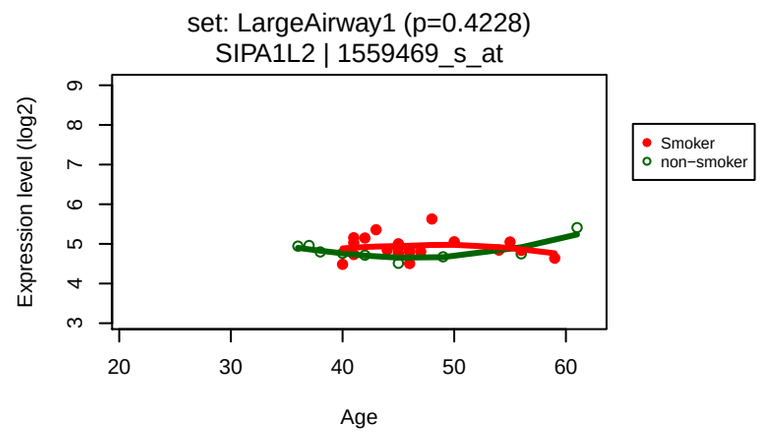

Gene expression of smokers and nonsmokers
gene: SIPA1L2 | 1559469_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0289
p=0.3579
p=0.2983
p=0.4228
p=0.8424
p=0.9247
p=0.2092
p=0.0118
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

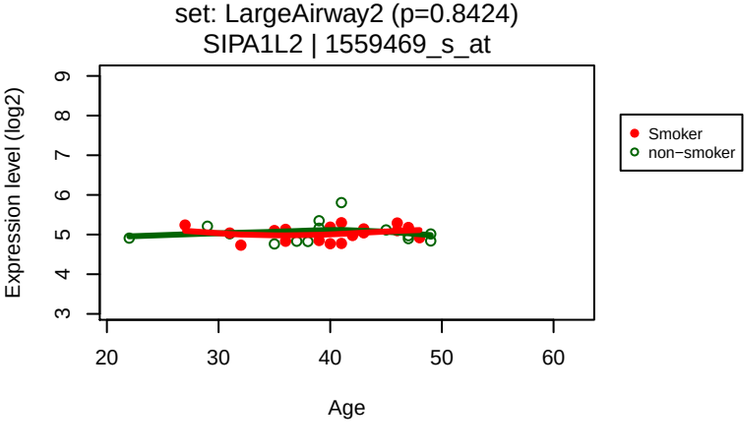

Gene expression ratio (smokers/nonsmokers)
gene: SIPA1L2 | 1559469_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0289
p=0.3579
p=0.2983
p=0.4228
p=0.8424
p=0.9247
p=0.2092
p=0.0118
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

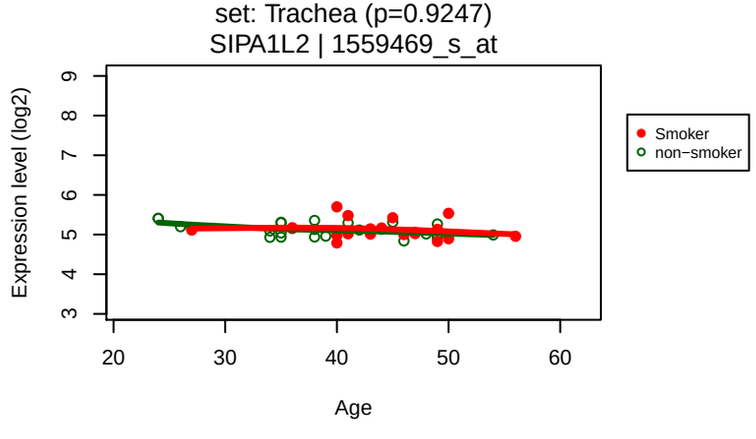

Gene expression ratio in Males
gene: SIPA1L2 | 1559469_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4765
p=0.4554
p=0.9124
p=0.0043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIPA1L2 | 1559469_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7728
p=0.5803
p=0.7535
p=0.9371
SmallAirway2
LargeAirway2
Trachea
Macrophage2

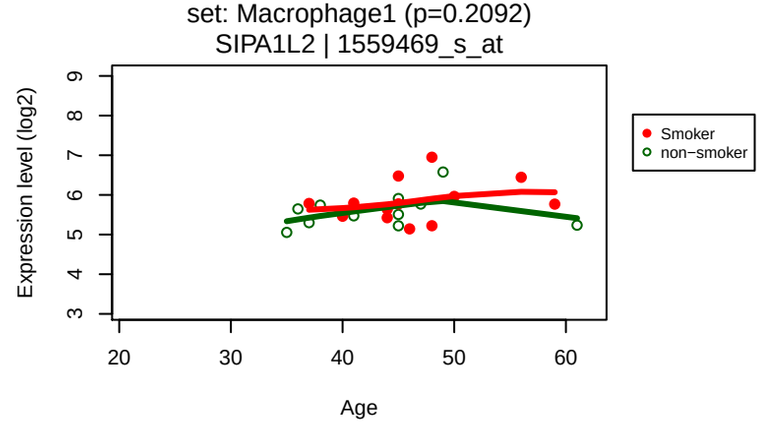

set: SmallAirway1 (p=0.0289)
SIPA1L2 | 1559469_s_at
Expression level (log2)
Age
Smoker
non−smoker

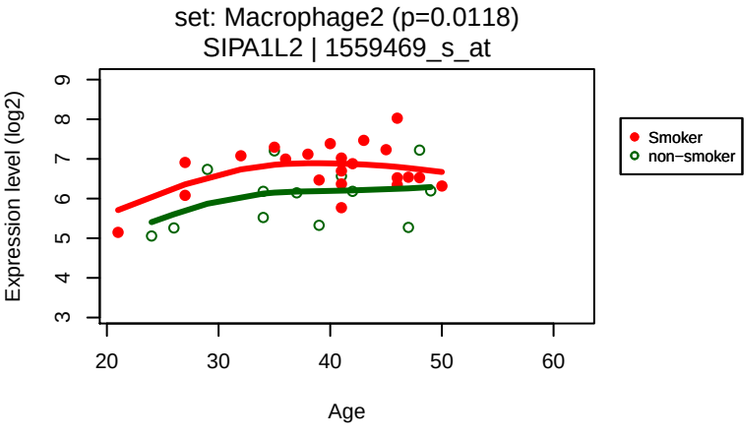

set: SmallAirway2 (p=0.3579)
SIPA1L2 | 1559469_s_at
Expression level (log2)
Age
Smoker
non−smoker

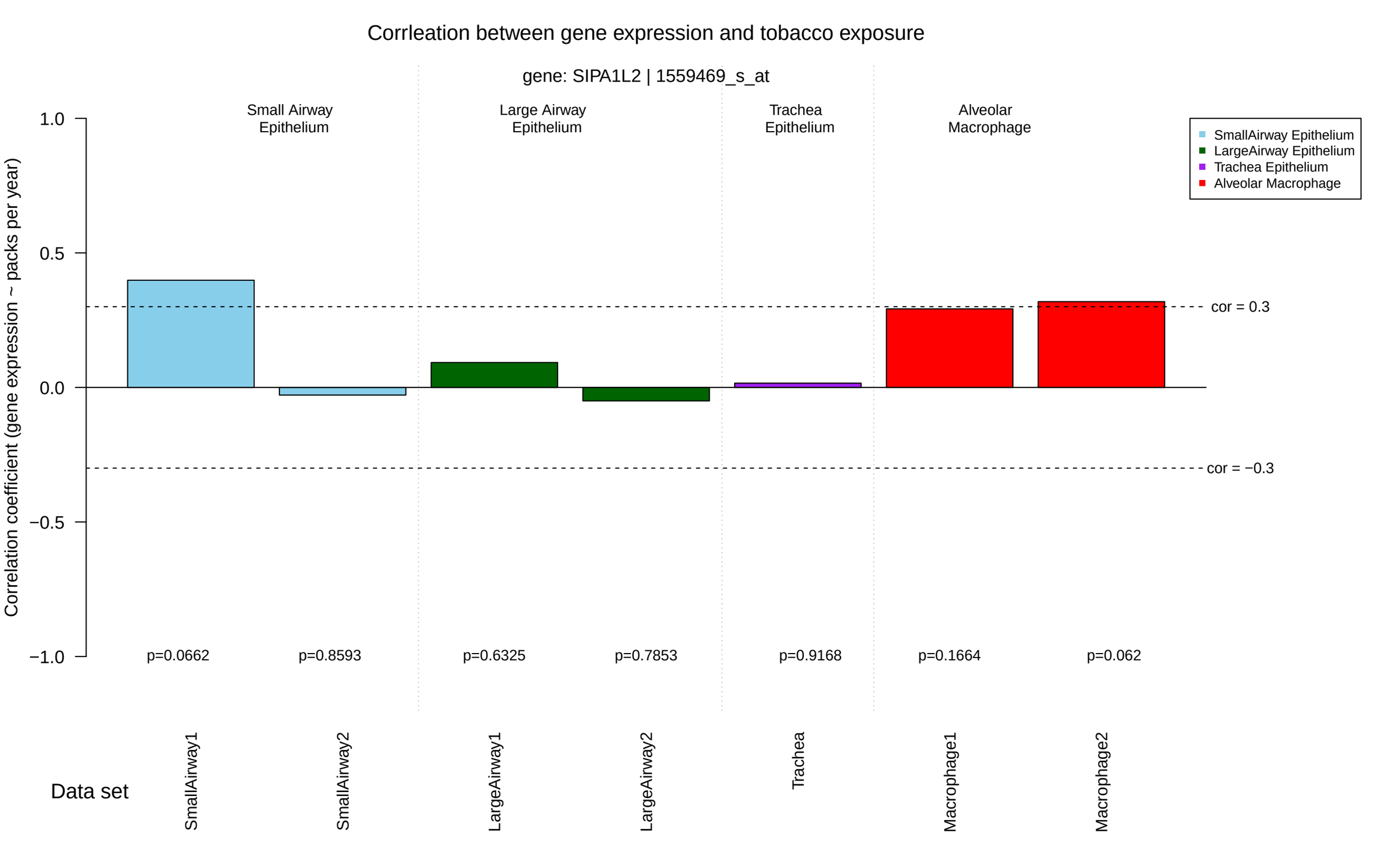

set: LargeAirway1 (p=0.4228)
SIPA1L2 | 1559469_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8424)
SIPA1L2 | 1559469_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9247)
SIPA1L2 | 1559469_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2092)
SIPA1L2 | 1559469_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0118)
SIPA1L2 | 1559469_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIPA1L2 | 1559469_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0662
p=0.8593
p=0.6325
p=0.7853
p=0.9168
p=0.1664
p=0.062
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2